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<u>Título/Title</u> Program CRIPCO User's Manual Version 2				
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<u>Resumen/Summary</u> The purpose of this technical report is to define the way to use the CRIPCO program, version 2. This program calculates the crippling allowable strain for stiffeners made of composite material. Considered stiffeners may be either riveted or cobonded or cocured to skin. Input data and output results are given by means of files.				
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REGISTRO DE REVISIONES/REVISIONS RECORD

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1.- Technical sheet of the program

Technical Note **ME-T-NT-100034**

PROGRAM

CRIPCO

REFERENCE

DATE

2010

OBJECTIVE

This program calculates the crippling allowable strain for stiffeners made of composite material.

LANGUAGE

Fortran 77

COMPUTER

Sun Solaris

NUMBER OF LINES (APROX.)

1030

EXECUTION MODE

Interactive

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VERSION

2.0

ADDITIONAL DOCUMENTS

Technical Note: NT-T-ADP-02002.
CRIPCO Program, v. 1.0, Theoretical Manual

2.- Introduction

This technical report describes the usage procedure for version 2.0 of CRIPCO program. Two examples are included.

This program calculates the crippling allowable strains for composite material stiffeners. Either riveted or cobonded or cocured stiffeners are considered.

Once the geometry and material data are introduced, the program calculates the local buckling strain and the crippling strain of each segment of the stiffener. This crippling strain is limited to the laminate allowable compression strain.

Only tow possible boundary conditions are considered for each segment: free-supported or supported-supported. Other conditions taking into account the stiffness supplied by the corners of the section to each segment is not considered.

The output results for this program are the local buckling microstrains and the crippling microstrains for each segment of the stiffener, as well as the stiffener crippling microstrains.

Differences between versions:

- Version 1.0:

Initial version developed in Fortran 77.

Input data are given interactively and the results are shown on the screen.

- Version 2.0:

The data flow of the program is modified. Now, the input data as well as the output results are provided by means of files:

- Input file: With the input data grouped in the same way as for other applications.
- Output file: With all the input data and output results in printable form.

The user can choose the language: Spanish or English.

The program is restructured to allow the creation of a dynamic link library DLL for Windows operative system.

In parallel with this, a Windows application has been developed as Graphic User Interface GUI for the program CRIPCO. This application uses the same input file and provides the same output results. The crippling analysis is performed by calling to the DLL, compiled from the same source as the official Fortran version.

3.- Execution of the program

Prior to the program execution, it is necessary to create the input data file, as described in section 4. The program can be then executed by typing its **full path name**.

If the software symbol table has been previously assigned (consult with the System Manager in order to know how to assign this symbol table), it can be executed by typing the command:

```
$scripco
```

The program asks for the language to use, the name of the input file and the name of the two output files.

```
Idioma / Language:
=====

 1 ... ESPAÑOL

 2 ... ENGLISH

Idioma / Language: 2

Name of the input file: Example.dat

Name of the output file: Example.txt
```

Details of the input and output files are provided in next sections. The CPU time for calculation is negligible.

It is possible to execute the program with a single command, including the input and output filenames and the language as parameters. The above example can be executed as follows:

```
$scripco Example.dat Example.txt
```

If the third parameter (language) is omitted, the Spanish language will be taken by the program.

4.- Input data file

This file contains all the necessary data that define the problem, as well as the different options required for solving it.

Basically, it contains four data groups, all of them are necessary. They will be numbered from 1 to 4 and must be in order.

All the numerical data contained in this file will be read in a free format. The variables will be separated by one or more blank spaces. The numbering of each group will be included in the first column.

Listed below are the contents of each data group:

GROUP 1: Title of the problem:

Four lines containing the title of the problem to be solved.

Line 1 -----> 1 First line of the title

Line 2 -----> Second line of the title

Line 3 -----> Third line of the title

Line 4 -----> Fourth line of the title

GROUP 2: Length of the stiffener:

This group has three lines with the following information

Line 1 -----> 2 “Optional commentary”

Line 2 -----> Length

Where Length = Spacing between supports for the stiffener (e.g.: ribs, frames)

GROUP 3: Number of segments:

The structure of this group is the following:

Line 1 -----> 3 “Optional commentary”

Line 2 -----> Ns

Where Ns = Number of segments in which the stiffener can be divided.

Several considerations must be taken into account for the segments adjacent to the panel.

- In the case the panel thickness is different at each side of the stiffener, the thickest panel will be extended until it completely covers the stiffener segment.

- If the stiffener is riveted to the panel, the segment is considered separately (see Figure 1:).

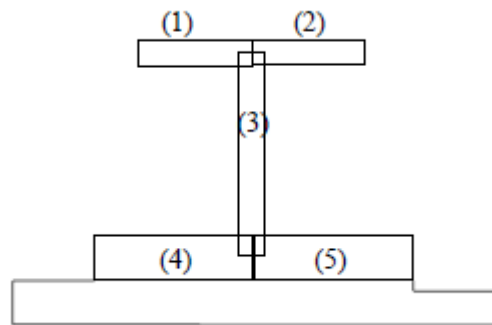


Figure 1: Stiffener riveted to the skin

- If the stiffener is cobonded or cocured to the skin, two possible configurations may take place:
 - If the overall thickness is equal or greater than twice the maximum skin thickness, $t_t \geq \max(t_{sk1}, t_{sk2})$, the skin is assumed to belong to the stiffener, making up a unique segment with it (see Figure 2: right side).
 - On the other hand, if the overall thickness is lower than twice the maximum skin thickness, $t_t < \max(t_{sk1}, t_{sk2})$, the stiffener segment is assumed to belong to the skin, so no segment is considered for crippling or local buckling calculation (see Figure 2: left side).

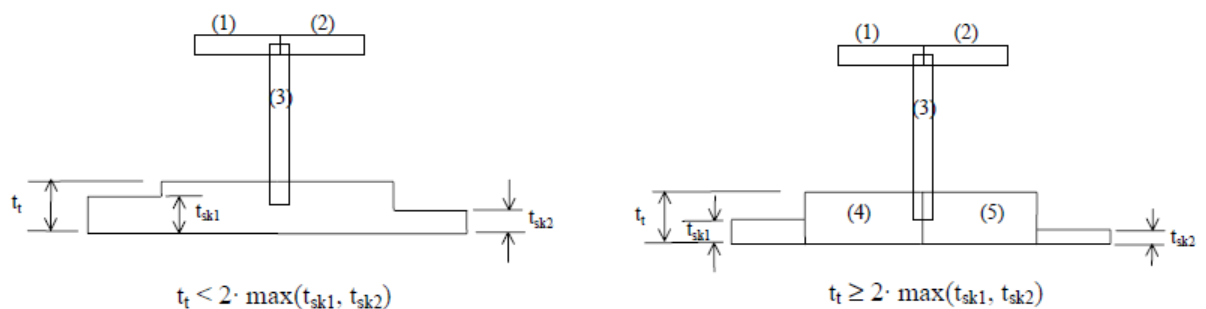


Figure 2: Stiffener cocured or cobonded to skin

GROUP 4: Segment data

This group has N_s+1 lines with the following content:

Line 1 -----> 4 “Optional commentary”

Line 2 -----> $I_p(1)$ $I_s(1)$ $W(1)$ $t(1)$ $E_c(1)$ $D_{11}(1)$ $D_{12}(1)$ $D_{22}(1)$ $D_{33}(1)$ $\epsilon_c(1)$

Line 3 -----> $I_p(2)$ $I_s(2)$ $W(1)$ $t(2)$ $E_c(2)$ $D_{11}(2)$ $D_{12}(2)$ $D_{22}(2)$ $D_{33}(2)$ $\epsilon_c(2)$

.....

.....

Line N_s-1 ---> $I_p(N_s)$ $I_s(N_s)$ $W(N_s)$ $t(N_s)$ $E_c(N_s)$ $D_{11}(N_s)$ $D_{12}(N_s)$ $D_{22}(N_s)$ $D_{33}(N_s)$ $\epsilon_c(N_s)$

Where: $I_p(i)$ = Orientation of the segment “i”. It is an integer with the following meaning:

$I_p(i) = 1$ If the segment is parallel to the skin.

$I_p(i) = 2$ If the segment is perpendicular to the skin.

$I_s(i)$ = Support condition for segment “i”. It is an integer with the following meaning:

$I_s(i) = 1$ If the segment has one side supported and the other free, as segments (1) and (2) in previous figures.

$I_s(i) = 2$ If the segment has both sides supported, as segment (3) in previous figures.

$W(i)$ = Is the width of the segment, measured along the middle line of the section.

$t(i)$ = Is the thickness of the segment. For stiffeners cocured or cobonded to the skin, the same considerations made for the number of segments definition must be observed.

$E_c(i)$ = Is the Young modulus under compression, corresponding to the mean properties of the laminate.

$D_{xy}(i)$ = Are the terms D_{11} , D_{12} , D_{22} and D_{33} of the flexural stiffness matrix $[D]$, under compression, corresponding to B-values of the laminate.

$\varepsilon_c(i)$ = Are the allowable microstrains under compression, corresponding to B-values of the laminate. It is the upper limit for the crippling microstrains of each segment.

5.- Output results

This section includes the description of the output result file as well as the possible error messages.

5.1 Output result file

This file provides the following information:

- **Summary of the input data:**
 - = Title.
 - = Stiffener Length
 - = Number of segment
 - = Segment data
- **Allowable microstrains for each segment:**
 - = Allowable microstrains for Local Buckling of the segment.
 - = Allowable microstrains for Crippling of the segment.
- **Allowable microstrains for the stiffener:**
 - = Allowable microstrains for Crippling of the stiffener.

5.2 Errors:

When the program detects an error in the input data or writing the results, it writes the corresponding message in the console and aborts the execution of the program.

The error messages that might appear in the console are listed below:

Error messages. English output:

ERROR: The input file cannot be opened

ERROR: Erroneous group number in the input file

ERROR: Group 1 missing in input file (Title)

ERROR: End of file while reading group 1 in the input file (Title)

ERROR: Group 2 missing in the input file (Length)

ERROR: Reading error while reading group 2 in the input file (Length)

ERROR: The length must be greater than 0

ERROR: Group 3 missing in input file (No. of segments)

ERROR: Reading error while reading group 3 in the input file (No. of segments)

ERROR: The number of segments must be greater than 0

ERROR: Group 4 missing in input file (Segment data)

ERROR: Reading error while reading group 4 in the input file (Segment data)

ERROR: Error in the segment position. It must be 1 or 2

ERROR: Error in the definition of the support condition. It must be 1 or 2

ERROR: The mid-line width of the segment must be greater than 0

ERROR: The thickness of the segment must be greater than 0

ERROR: The mean value of E_c for the laminate must be greater than 0

ERROR: The allowable value D_{11} for the laminate must be greater than 0

ERROR: The allowable value D_{12} for the laminate must be greater than 0

ERROR: The allowable value D_{22} for the laminate must be greater than 0

ERROR: The allowable value D_{33} for the laminate must be greater than 0

ERROR: The allowable microstrains in compression E_{cu} for the laminate must be greater than 0

ERROR: The output file cannot be opened

ERROR: Writing error in the output file

Error messages. Spanish output:

ERROR: No se puede abrir el fichero de entrada

ERROR: Número de grupo erróneo en el fichero de entrada

ERROR: Grupo 1 ausente en el fichero de entrada (Título)

ERROR: Fin de fichero mientras se leía el grupo 1 en el fichero de entrada (Título)

ERROR: Grupo 2 ausente en el fichero de entrada (Longitud)

ERROR: Error de lectura mientras se leía el grupo 2 en el fichero de entrada (Longitud)

ERROR: La longitud debe ser mayor que 0

ERROR: Grupo 3 ausente en el fichero de entrada (Nº de tramos)

ERROR: Error de lectura mientras se leía el grupo 3 en el fichero de entrada (Nº de tramos)

ERROR: El número de tramos debe ser mayor que 0

ERROR: Grupo 4 ausente en el fichero de entrada (Datos de tramo)

ERROR: Error de lectura mientras se leía el grupo 4 en el fichero de entrada (Datos de tramo)

ERROR: Error en la posición del tramo. Debe ser 1 ó 2

ERROR: Error en la definición de la condición de apoyo. Debe ser 1 ó 2

ERROR: La anchura media del tramo debe ser mayor que 0

ERROR: El espesor del tramo debe ser mayor que 0

ERROR: El valor medio de Ec para el laminado debe ser mayor que 0

ERROR: El valor admisible D11 para el laminado debe ser mayor que 0

ERROR: El valor admisible D12 para el laminado debe ser mayor que 0

ERROR: El valor admisible D22 para el laminado debe ser mayor que 0

ERROR: El valor admisible D33 para el laminado debe ser mayor que 0

ERROR: Las microdef admisibles a compresión Ecu para el laminado deben ser mayores que 0

ERROR: No se puede abrir el fichero de salida

ERROR: Error de escritura en el fichero de salida

6.- Windows application

An application has been developed in Visual Basic for Windows environment. The original routines of the program for input data, calculation and output results have been included in a Dynamic Link Library, which is used by this application. Therefore the same results will be obtained, with only differences due to rounding.

Since this is a Windows application, it must be installed in the user's PC before it can be executed. For installation, please consult with the system manager,

The application can be executed by "double click" in the file **Cripco.exe**, located in the PC's server, or by means of a **link** in the "Desktop" or in the "Start Menu".

The input data can be retrieved from an input data file or directly typed in the application form. The form has the following appearance:

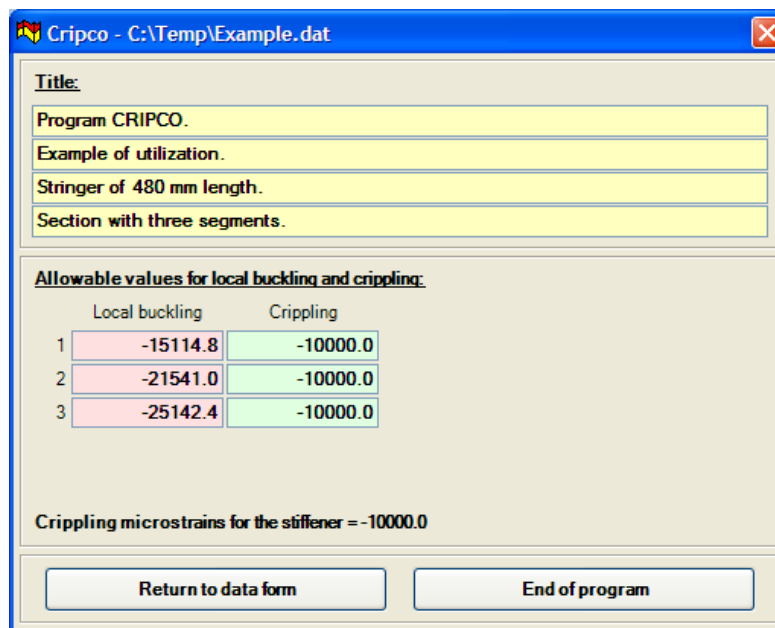
The screenshot shows the 'Cripco - C:\Temp\Example.dat' window. It contains several sections for data entry:

- Title:** Four text boxes containing 'Program CRIPCO.', 'Example of utilization.', 'Stringer of 480 mm length.', and 'Section with three segments.'
- Stiffener geometry:**
 - Stiffener Length: 480.0
 - No. of segments: 3
 - A table with 3 rows and 10 columns: Orient., Supp., Width, Thickness, Ec, D11, D12, D22, D33, Ecu.
- Output results file:**
 - Filename: C:\Temp\Example.txt
- Input file:** Three icons for file operations.
- Language:** Radio buttons for 'Español' and 'English' (selected), with a UK flag icon.
- Program execution:** A large button labeled 'Program execution'.

	Orient.	Supp.	Width	Thickness	Ec	D11	D12	D22	D33	Ecu
1	Parallel	1 Support	14.10	3.240	54932.1	148333.6	38753.2	155744.4	44147.6	10000.0
2	Normal	2 Supports	18.92	1.800	62374.4	19899.7	5186.2	35157.4	6111.1	10000.0
3	Parallel	1 Support	5.10	1.800	62374.4	19899.7	5186.2	35157.4	6111.1	10000.0

Figure 3: Input data form.

The program can be executed clicking the button “Program execution”. Then the application solves the problem, writes the output files and shows the results in the following form:



The screenshot shows a software window titled "Cripco - C:\Temp\Example.dat". It contains a "Title:" section with the following text: "Program CRIPCO.", "Example of utilization.", "Stringer of 480 mm length.", and "Section with three segments.". Below this is a section titled "Allowable values for local buckling and crippling:" which contains a table with two columns: "Local buckling" and "Crippling". The table has three rows of data. At the bottom of the table section, it states "Crippling microstrains for the stiffener = -10000.0". At the very bottom of the window are two buttons: "Return to data form" and "End of program".

	Local buckling	Crippling
1	-15114.8	-10000.0
2	-21541.0	-10000.0
3	-25142.4	-10000.0

Crippling microstrains for the stiffener = -10000.0

Return to data form End of program

Figure 4: Output results form.

7.- Examples

Input data and obtained results are presented for the two following examples.

7.1 Example 1

The case to solve is a T-shape stringer cobonded or cocured to the skin. The geometry is as shown on next figure.

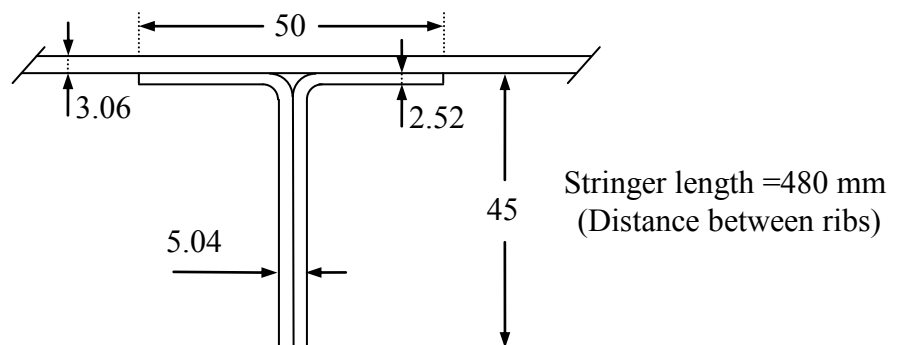


Figure 5: Example 1 geometry.

The stringer material is:

- Tape of group V:

The lamina properties under compression for this material are the following:

	E_{11}	E_{22}	G_{12}	μ_{12}	ϵ_c
Mean values	122000	8900	4700	0.3	---
Allowable values	120000	8600	4500	---	10000

- Lay-up: Stringer..... ((+45/-45/0/0/0/90/0)S
Skin(45/-45/0/-45/45/90/-45/+45/0)\$

The skin thickness and the total thickness of the stringer foot plus the skin are:

$$t_{sk} = 3.06 \text{ mm}$$

$$t_t = 2.52 + 3.06 = 5.58 \text{ mm}$$

Since $t_t < 2 \cdot t_{sk}$, only the vertical segment is going to be considered.

This segment is perpendicular to skin, $I_p = 2$, and it has one end free and the other end supported, $I_s = 1$.

The dimensions to be considered for this segment are:

$$W = 15 + 1.06 - 1.58/2 = 15,27 \text{ mm}$$

$$t = 1,04 \text{ mm}$$

The material properties required by the program for the laminate of the considered segment can be calculated from the above data (i.e.: by running ANCOLAB program). The results are the following:

$$E_c = 79410.4 \text{ N/mm}^2$$

$$D_{11} = 825340.3 \text{ N/mm}^2$$

$$D_{12} = 133448.9 \text{ N/mm}^2$$

$$D_{22} = 344068.0 \text{ N/mm}^2$$

$$D_{33} = 153754.0 \text{ N/mm}^2$$

$$\varepsilon_c = 10000.0 \text{ } \mu\varepsilon$$

The input and output files corresponding to this example are shown on next pages:

Input file:

```

1 Program CRIPCO.
  Example of utilization number 1
  Stringer of 480 mm length.
  Section with only one segment.
2 Length
  480.0
3 Number of segments
  1
4 Segment data: Orient.  Apoyo  Anchura  Espesor  Ec  D11  D12  D22  D33  Ecu
  2  1  45.27  5.04  79410.4  825340.3  133448.9  344068.0  153754.0  10000.0

```

Output file

```
*****
*
*          <<< CRIPCO >>>
*
*****
* Calculation of the crippling allowable *
* value for composite stringers        *
*****
```

Title:

=====

Program CRIPCO.
Example of utilization number 1.
Stringer of 480 mm length.
Section with only one segment.

Input data:

=====

Length of the stiffener: 480.0

Number of stiffener segments: 1

Segment	Width	Thickness	Orient.	Supp.	Ec	D11	D12	D22	D33	Ecu
1	45.27	5.040	2	1	79410.4	825340.3	133448.9	344068.0	153754.0	10000.0

Allowable values for local buckling and crippling:

=====

Segment	Local buckling	Crippling
1	-2602.8	-5765.1

```
*****
* Crippling microstrains for the stiffener: -5765.1 *
*****
```

7.2 Example 2

The case to solve is a Z-shape stringer cobonded or cocured to the skin. The geometry is as shown on next figure.

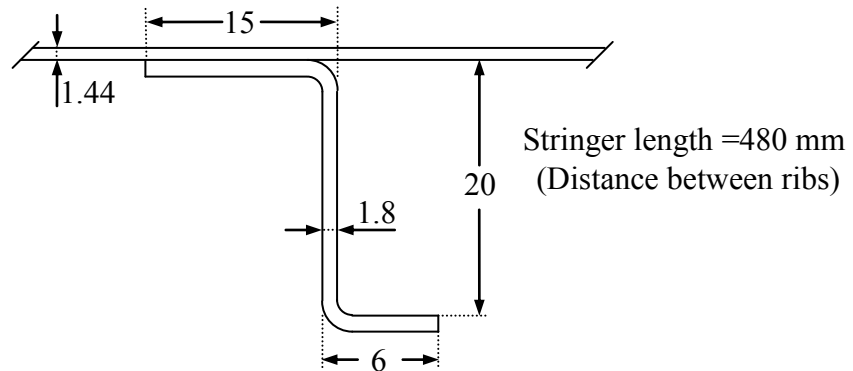


Figure 6: Example 2 geometry.

The stringer material is:

- Tape of group V:

The lamina properties under compression for this material are the following:

	E_{11}	E_{22}	G_{12}	μ_{12}	ε_c
Mean values	122000	8900	4700	0.3	---
Allowable values	120000	8600	4500	---	10000

- Lay-up: Stringer..... (90/45/0/0/-45)S

Skin(90/45/0/-45)S

The skin thickness and the total thickness of the stringer foot plus the skin are:

$$t_{sk} = .44 \text{ mm}$$

$$t_t = .8 + .44 = 1.24 \text{ mm}$$

Since $t_t > 2 \cdot t_{sk}$, three segments are going to be considered.

The upper and lower flanges are parallel to the skin, $I_p = 1$, and they have one end free and the other supported, $I_s = 1$. The web of the stringer is perpendicular to the skin, $I_p = 2$ and it has both ends supported, $I_s = 2$. The dimensions to be considered for the three segments are:

$$W_{upper} = .5 - .8/2 = .1 \text{ mm}$$

$$t_{upper} = .8 + .44 = 1.24 \text{ mm}$$

$$W_{web} = (1.44 - 1.8 + 1.24)/2 = 8,92 \text{ mm}$$

$$t_{web} = ,8 \text{ mm}$$

$$W_{lower} = 5 - ,8/2 = 5,1 \text{ mm}$$

$$t_{lower} = ,8 \text{ mm}$$

The material properties required by the program for the laminate can be calculated from the above data (i.e.: by running ANCOLAB program). The results are the following:

Upper flange + Skin

$$E_c = 54932.1 \text{ N/mm}^2$$

$$D_{11} = 148333.6 \text{ N/mm}^2$$

$$D_{12} = 38753.2 \text{ N/mm}^2$$

$$D_{22} = 155744.4 \text{ N/mm}^2$$

$$D_{33} = 44147.6 \text{ N/mm}^2$$

$$\varepsilon_c = 10000.0 \text{ } \mu\varepsilon$$

Lower flange and web

$$E_c = 62374.4 \text{ N/mm}^2$$

$$D_{11} = 19899.7 \text{ N/mm}^2$$

$$D_{12} = 5186.2 \text{ N/mm}^2$$

$$D_{22} = 35157.4 \text{ N/mm}^2$$

$$D_{33} = 6111.1 \text{ N/mm}^2$$

$$\varepsilon_c = 10000.0 \text{ } \mu\varepsilon$$

The input and output files corresponding to this example are shown on next pages:

Input file:

```

1 Program CRIPCO.
  Example of utilization number 2
  Stringer of 480 mm length.
  Section with three segments.
2 Length
  480.0
3 Number of segments
  3
4 Segment data: Orient.  Apoyo  Anchura  Espesor  Ec  D11  D12  D22  D33  Ecu
  1  1  14.10  3.24  54932.1  148333.6  38753.2  155744.4  44147.6  10000.0
  2  2  18.92  1.80  62374.4  19899.7  5186.2  35157.4  6111.1  10000.0
  1  1   5.10  1.80  62374.4  19899.7  5186.2  35157.4  6111.1  10000.0

```

Output file

```
*****
*
*          <<< CRIPCO >>>
*
*****
* Calculation of the crippling allowable
* value for composite stringers
*****
```

Title:

=====

Program CRIPCO.
Example of utilization number 2.
Stringer of 480 mm length.
Section with three segments.

Input data:

=====

Length of the stiffener: 480.0

Number of stiffener segments: 3

Segment	Width	Thickness	Orient.	Supp.	Ec	D11	D12	D22	D33	Ecu
1	14.10	3.240	1	1	54932.1	148333.6	38753.2	155744.4	44147.6	10000.0
2	18.92	1.800	2	2	62374.4	19899.7	5186.2	35157.4	6111.1	10000.0
3	5.10	1.800	1	1	62374.4	19899.7	5186.2	35157.4	6111.1	10000.0

Allowable values for local buckling and crippling:

=====

Segment	Local buckling	Crippling
1	-15114.8	-10000.0
2	-21541.0	-10000.0
3	-25142.4	-10000.0

```
*****
* Crippling microstrains for the stiffener: -10000.0 *
*****
```